

How private is your attention?

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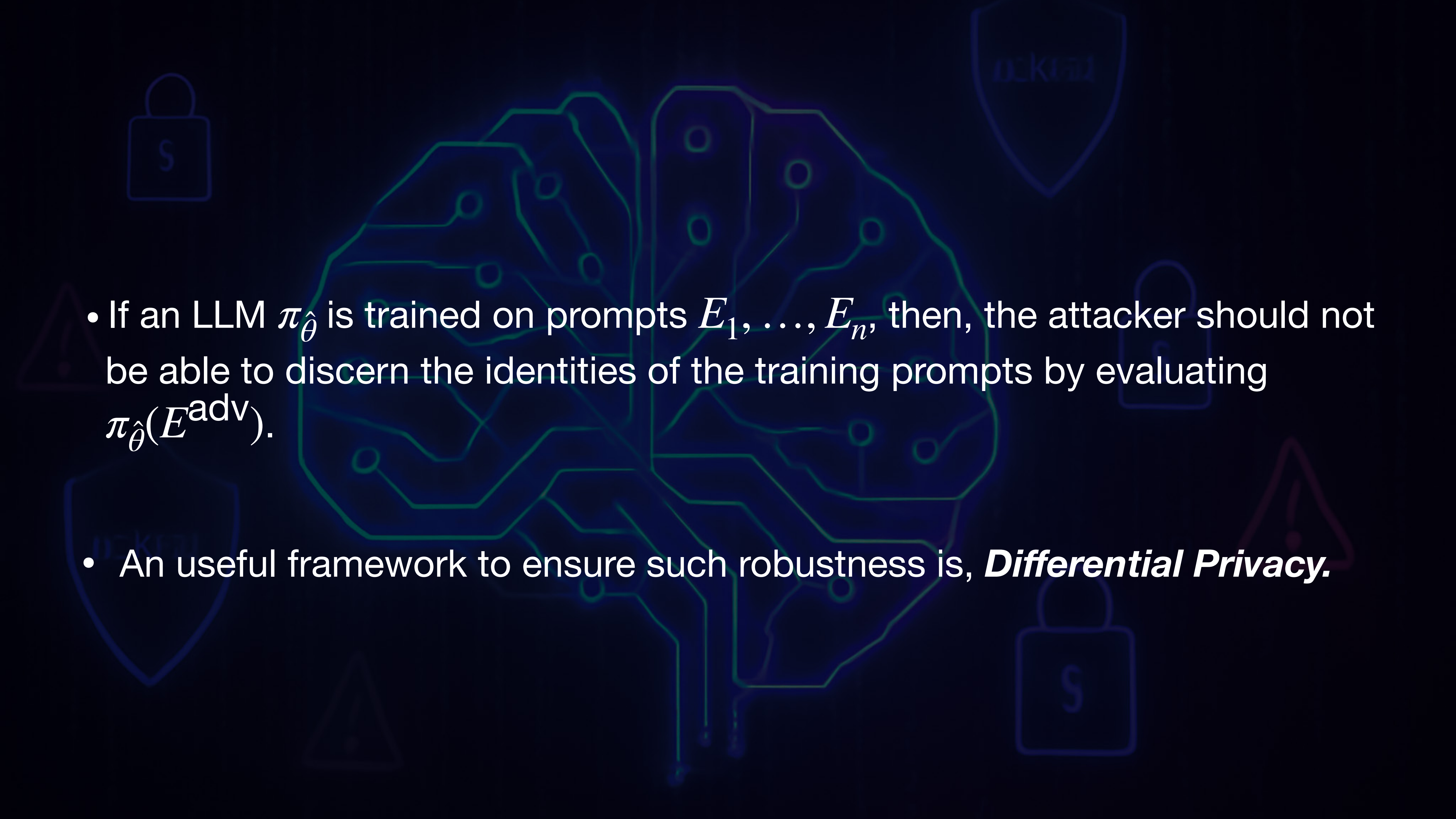
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**Adversarial
Prompting**



Output

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- If an LLM $\pi_{\hat{\theta}}$ is trained on prompts $E_1, \dots, E_n$, then, the attacker should not be able to discern the identities of the training prompts by evaluating $\pi_{\hat{\theta}}(E^{\text{adv}})$.
 - An useful framework to ensure such robustness is, ***Differential Privacy***.

Differential Privacy

An LLM π is ε, δ - differentially private, if, for any two sets of prompts $M = (E_1, \dots, E_n), M' = (E'_1, \dots, E'_n)$, differing in at most one entry, it holds that $P[\pi(M) \in A] \leq e^\varepsilon P[\pi(M') \in A] + \delta$, for all sets A .

- Changing one training prompt, by however much, should not be reflected at all in the output.
- For example, $\bar{X}$ is not differentially private, since changing one of the X_i ' can change the mean by a significant amount.

How to induce differential privacy?

Two key ideas:

- *Projection:*

If we project the output onto a compact space, the attacker would not be able to discern changes in the training prompts.

- *Addition of calibrated noise:*

To maintain the randomness of the output, we add noise along with projection. Too much noise leads to loss of accuracy, so the noise must be calibrated.

Making Transformers differentially private

- Consider a single linear attention head $f(E; \theta) = E + W^{PV}E \cdot \frac{E^\top W^{KQ}E}{L}$,
where, $E = \begin{pmatrix} x_1 & x_2 & \cdots & x_L & x_{L+1} \\ y_1 & y_2 & \cdots & y_L & 0 \end{pmatrix}$ is the input prompt. The final output is $\hat{y}_{L+1}(E) = (f(E; \theta))_{(D+1, L+1)}$; D is the dimension of x_i 's.
- Also known as In-context learning, as pre-trained transformers can predict accurately for different models without further fine-tuning.

Making Transformers differentially private

- Importantly, it can be argued $\hat{y}_{L+1} = \langle \Gamma, Z \rangle$, where Γ depends on the key-query matrices, and Z depends on E .
- Given training prompts $E_1, \dots, E_N$, we train the LLM as:

$$\mathcal{L}_\lambda(\Gamma) := \frac{1}{N} \sum_{i=1}^N \left(y_{i,L+1} - \langle \Gamma, Z_i \rangle \right)^2 + \lambda \|\Gamma\|_F^2.$$

Now this becomes a ridge-regression. So we can use gradient descent. Instead of projection and noise addition on the final output, we perform these on each step of gradient descent.

DP-pretraining

$$\mathcal{L}_\lambda(\Gamma) := \frac{1}{N} \sum_{i=1}^N \left(y_{i,L+1} - \langle \Gamma, Z_i \rangle \right)^2 + \lambda \|\Gamma\|_F^2.$$

At t -th step :

- Generate noise w_t with appropriate variance.
- Update

$$\Gamma^{t+1} \leftarrow \Pi_R \left((1 - 2\lambda\eta)\Gamma^t - \eta N^{-1} \sum_{k=1}^N \left(\langle \Gamma^t, \tilde{Z}_k \rangle - \text{clip}(y_{k,L+1}) \right) \tilde{Z}_k + w_t \right)$$

Here: Learning-rate

A projection of Z_i matrices

We finally have an umbrella projection onto R .

Cost of Privacy and Early Stopping

- If $\hat{\Gamma}$ is the fitted Γ based on training prompts, and Γ^* is the globally optimal Γ , then

Cost of privacy: $(\langle \hat{\Gamma}, Z \rangle - \langle \Gamma^*, Z \rangle)^2$ for Z from a test prompt E .

Focus on the high-dimensional regime: $N \gtrsim D^2 L^{-2}, D^2 \gtrsim \log(NL)$.

If the number of gradient-descent iterations $T \asymp \log r$ for some r , then with high pr.,

$$\text{Cost of privacy} \lesssim \frac{N \log^3(NL)}{L^2 \lambda^2} \left(\frac{1}{r} + \frac{D \log^3 r}{N^3} \left(1 + \log^2(NL) \frac{N}{L^2 \lambda^2} \right) \frac{\log(1/\delta)}{\varepsilon^2} \right)$$

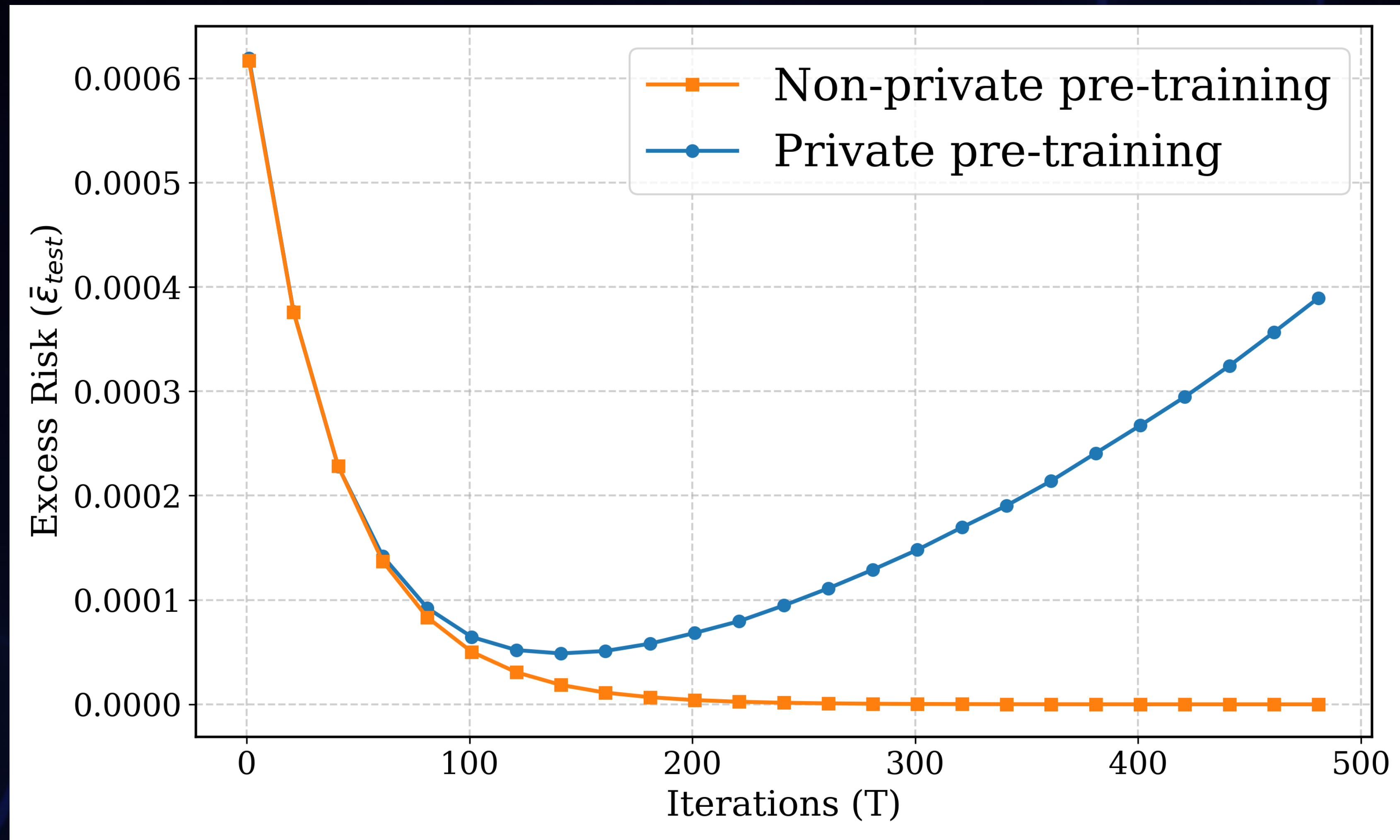
Decreases with T

(Error due to gradient descent)

Increases with T

(Error due to noise injection)

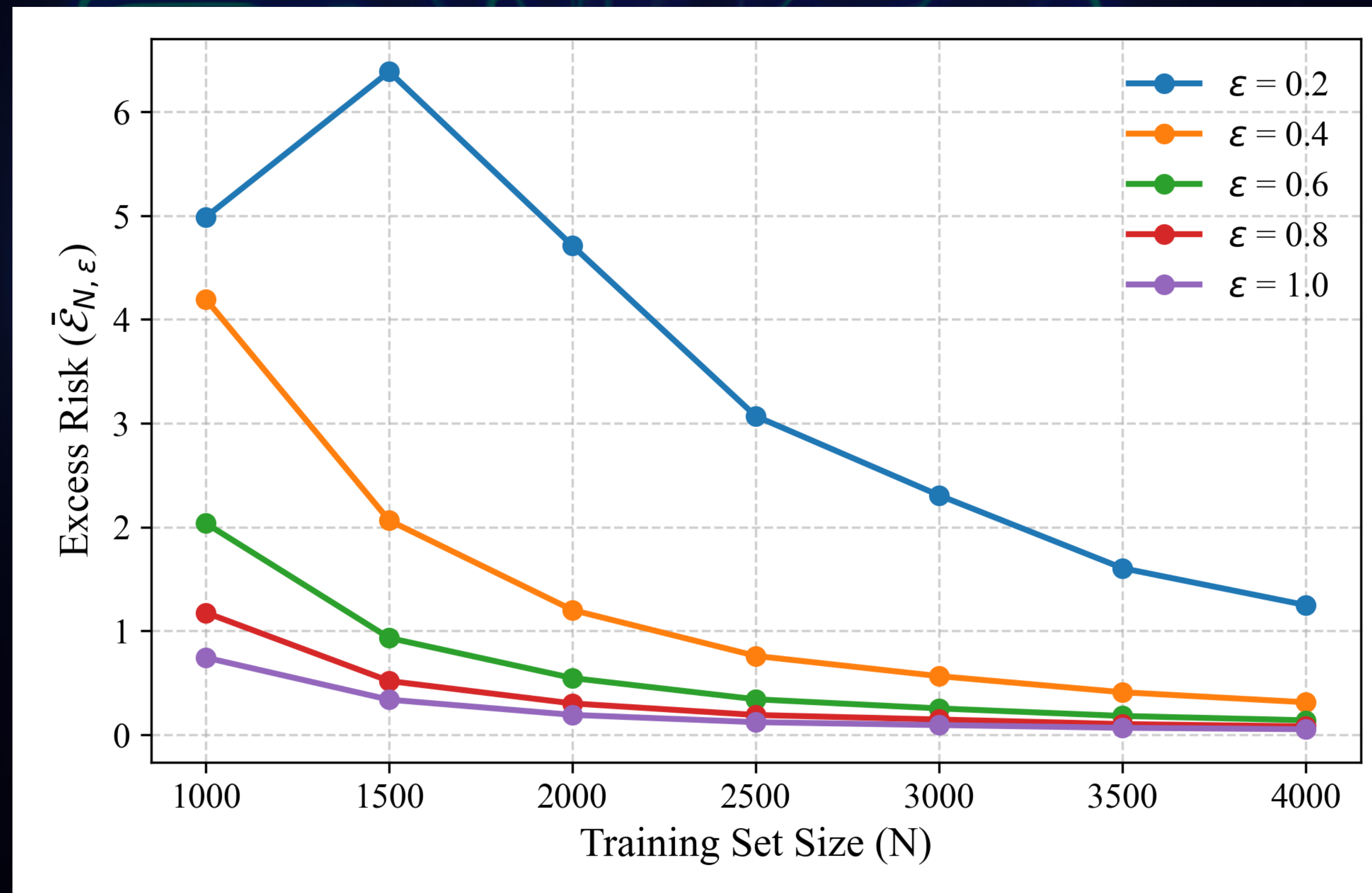
DP parameters



In DP-pretraining, accuracy decreases if T is too high; so we need ***Early Stopping***.

When $N/D^2 = O(1)$, $L/D = O(1)$, the cost of privacy scales as

Cost of Privacy or excess risk $\lesssim \frac{D^2}{N^2 L^2} \frac{\log(1/\delta)}{\epsilon^2}$, with $T \asymp \log N$, $\eta \asymp 1$.



Robustness properties

Consider adversary replaces training prompt E_i by

$$E_{\text{bad},i}(\mu, \alpha) = \begin{pmatrix} x'_{i,1} & x'_{i,2} & \cdots & x'_{i,L} & x'_{i,L+1} \\ y'_{i,1} & y'_{i,2} & \cdots & y'_{i,L} & 0 \end{pmatrix} \in \mathbb{R}^{(D+1) \times (L+1)}, \text{ where,}$$
$$x'_{i,k} = x_{i,k} + \mu, y'_{i,\ell} = y_{i,\ell} + \alpha.$$

Let $\hat{\Gamma}, \hat{\Gamma}_{\text{bad}}$ be the DP-pretrained estimates based on unperturbed and perturbed training prompts. Let $\Gamma^*, \Gamma_{\text{bad}}^*$ be the corresponding globally optimal parameters (for the ridge-representation).

- If μ, α are neither too big nor too small, then it happens that

$$(\langle \hat{\Gamma}, Z \rangle - \langle \hat{\Gamma}_{\text{bad}}, Z \rangle)^2 \xrightarrow{p} 0, \text{ but, } (\langle \Gamma^*, Z \rangle - \langle \Gamma_{\text{bad}}^*, Z \rangle)^2 \xrightarrow{p} \infty.$$

Takeaways

- Maintaining training privacy is important.
- First characterization of DP for in-context learning of transformers.
- Projection + Calibrated Noise achieves differential privacy.
- Early stopping guarantees optimal accuracy on per with non-private training, While maintaining privacy.
- Differentially private algorithm has in-built robustness even against attacks on training procedure.

Questions??

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